

MESQUITE, TX, USA

WMO: 720291

Lat: 32.747N

Lon: 96.531W

Elev: 136

StdP: 99.70

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-2.5	-12.1	1.3	0.0	-9.6	1.7	2.0	11.8	13.7	10.8	12.8	3.2	330	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.4	(c) 37.8	(d) 22.9	(e) 37.1	(f) 22.9	(g) 35.5	(h) 23.0	(i) 25.7	(j) 31.9	(k) 25.2	(l) 31.6	(m) 24.7	(n) 31.1	(o) 3.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.2	28.1	23.0	18.1	27.1	22.8	17.8	27.0	79.8	31.0	77.4	31.5	75.4	31.1	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.0	(c) 8.1	(e) -7.7	(f) 40.0	(g) 2.7	(h) 1.5	(i) -9.6	(j) 41.0	(k) -11.2	(l) 41.9	(m) -12.8	(n) 42.8	(o) -14.8	(p) 43.9		
(4) 10.5	(5) 9.0	(6) 8.1	(7) -7.7	(8) 40.0	(9) 2.7	(10) 1.5	(11) -9.6	(12) 41.0	(13) -11.2	(14) 41.9	(15) -12.8	(16) 42.8	(17) -14.8	(18) 43.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	7.9	9.6	14.4	18.2	22.5	27.2	28.6	29.0	25.0	19.1	13.9	8.3
	DBStd	8.65	5.46	5.78	5.02	4.29	3.71	2.01	1.99	2.62	3.19	4.37	4.93	5.46
	HDD10.0	316	106	70	18	2	0	0	0	0	1	18	101	
	HDD18.3	1286	326	247	139	53	12	0	0	0	1	44	148	316
	CDD10.0	3483	41	59	155	247	388	515	575	589	450	281	135	47
	CDD18.3	1413	2	3	18	49	142	265	317	331	201	67	16	3
	CDH23.3	15638	12	26	109	384	1261	2952	3825	4206	2123	611	109	21
(13)	CDH26.7	7607	1	1	11	93	444	1444	2046	2362	1008	177	17	2
(14)	Wind	WSAvg	3.4	3.5	3.9	4.4	4.4	3.6	3.2	2.6	2.5	2.9	3.4	3.5
(15) Precipitation	PrecAvg	975	69	67	90	93	123	92	51	56	71	109	79	75
	PrecMax	1683	159	187	179	185	386	273	125	129	237	298	258	221
	PrecMin	530	5	6	2	30	15	17	0	0	12	7	12	6
	PrecStd	254	41	45	48	46	71	66	37	35	55	77	59	50
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	27.0	27.8	32.0	33.8	37.2	38.7	40.2	37.4	32.7	28.8	26.2
		MCWB	16.0	16.9	18.3	21.6	22.2	22.5	23.3	23.1	21.6	21.8	19.3	17.0
	2%	DB	22.4	23.0	26.2	28.9	32.4	35.9	37.4	38.7	35.7	30.9	26.3	22.4
		MCWB	15.2	14.9	17.7	20.5	22.2	22.5	23.3	22.9	21.8	22.0	18.3	16.6
	5%	DB	20.1	21.4	24.1	27.4	31.1	34.8	36.2	37.5	33.8	28.0	23.9	20.1
		MCWB	15.5	15.4	17.3	19.4	22.3	22.9	23.2	22.9	21.7	19.6	17.9	15.2
	10%	DB	17.6	18.9	22.5	25.9	29.1	32.9	34.9	36.2	32.4	27.2	22.3	17.6
		MCWB	13.3	14.2	16.4	18.7	21.6	23.1	23.4	23.2	21.4	19.2	17.1	13.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.0	20.8	23.0	25.1	26.3	26.3	26.6	25.3	24.4	21.7	19.6
		MCDB	20.6	23.6	25.6	29.2	30.1	32.0	32.1	32.4	31.2	28.6	25.5	23.2
	2%	WB	18.0	17.9	19.5	21.8	24.0	25.4	25.5	25.6	24.6	23.5	20.1	18.0
		MCDB	20.2	21.4	23.6	26.8	29.4	31.0	31.9	32.5	30.5	27.6	24.0	20.9
	5%	WB	16.5	16.7	18.8	20.8	23.3	24.8	25.0	25.1	23.8	22.4	19.0	16.7
		MCDB	19.2	19.8	22.7	25.1	28.6	30.6	31.6	32.1	29.6	26.2	22.8	19.0
	10%	WB	13.7	15.1	17.8	20.0	22.5	24.2	24.6	24.5	23.1	21.3	17.8	14.4
		MCDB	17.3	18.4	21.1	24.3	27.7	30.3	31.0	31.6	28.7	24.9	20.8	17.4
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.4	11.8	12.0	11.0	11.2	11.6	12.4	12.2	12.7	12.0	11.5
		MCDBR	15.4	14.4	12.8	12.7	12.5	13.0	13.3	14.4	14.3	13.9	13.5	14.9
		MCWBR	9.7	8.9	7.0	6.3	5.0	3.9	3.4	3.5	4.9	6.6	7.8	9.6
	5% WB	MCDBR	12.0	12.2	10.9	10.7	10.3	10.5	11.0	11.9	11.2	11.2	11.2	11.9
		MCWBR	9.3	8.7	7.0	6.4	5.2	4.2	3.5	3.9	4.5	6.4	7.8	9.2
(40) Clear-Sky Solar Irradiance	taub		0.317	0.333	0.360	0.398	0.429	0.442	0.445	0.447	0.402	0.360	0.341	0.322
	taud		2.496	2.462	2.403	2.316	2.289	2.286	2.317	2.295	2.370	2.473	2.470	2.486
	Ebn at Noon		903	922	918	892	863	847	842	837	863	880	867	875
	Edh at Noon		87	99	113	129	134	134	129	130	116	97	88	83
(44) All-Sky Solar Radiation	RadAvg		2.82	3.27	4.37	5.33	5.76	6.49	6.63	6.11	5.14	4.03	3.01	2.49
	RadStd		0.28	0.52	0.35	0.39	0.49	0.61	0.45	0.41	0.45	0.60	0.38	0.32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page